

This is the first problem set.

Will not be marked or used in your continuous assessment.

But you will probably have trouble with later problem sets if you don't work on this one.

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1. Electric Dipole.

A positive charge q is placed at position $(0, +d/2)$, and a negative charge $-q$ is placed at position $(0, -d/2)$. This is an *electric dipole*.

- Look up the definition of the *dipole moment*. What is the magnitude of the dipole moment? What is the direction of the dipole moment?
- Calculate the electric field at the point $(x_0, 0)$ on the x -axis, by calculating the fields due to each charge and adding them. This would be vector addition. Make sure to draw pictures for yourself, showing directions and angles.
- Calculate the electric potential $V(x, y, z)$ at an arbitrary point $(x, y, 0)$ on the $z = 0$ plane, by scalar addition of the contributions from the two charges. Now try this for arbitrary z .
- Use your expression for V to calculate the electric field at point $(x_0, 0)$, using the relationship between $\mathbf{E}$ and V . How does this compare with the direct calculation of $\mathbf{E}$?

2. Line (thin rod) of continuous charge.

Consider a uniformly charged rod of length L , lying along the y -axis; the endpoints are at $(0, -L/2)$ and $(0, +L/2)$ and the center is at the origin. The linear charge density (charge per unit length) is λ .

- Do draw a picture of the situation. What is the total charge on the rod?
- Consider the point $\mathbf{P}$ on the x -axis, with coordinates $(x_0, 0)$. We want to calculate the electric field at this point. By symmetry, which direction should the field point towards?
- Take an infinitesimal element of the rod, between point $(0, y)$ and point $(0, y + dy)$. What is the charge within this element? What is the electric field contribution at point $\mathbf{P}$ due to this infinitesimal charge element? Evaluate separately the x - and y -components, dE_x and dE_y . Do these depend on the position y of the charge element?

- (d) You should now be able to calculate the total electric field components by adding up contributions from all infinitesimal elements, which means integrating from $y = -L/2$ to $y = +L/2$.
- (e) Does the expression simplify in the case $x_0 \gg L$? Is the expression similar to the expression for a point charge? Would you have expected this?
- (f) Now imagine that the rod is infinitely long, i.e., $L \rightarrow \infty$. By taking the limit of the expression derived for finite L , calculate the electric field in this limit. Could you also get this result directly by integrating from $y = -\infty$ to $y = +\infty$?
- (g) The electric field for an infinite line of charge (if you did this correctly), is inversely proportional to x_0 , not to the square of x_0 . The decrease of electric field with increasing distance is *slower* than compared to the electric field due to a single charge. Would you have expected this?
- (h) Going back to the finite case (rod of length L), calculate the electric potential due to an infinitesimal element. Then integrate to calculate the total electric potential due to the rod of charge.
- (i) Try taking the limit $L \rightarrow \infty$ for the electric potential. Is there a problem?